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Magnetostratigraphy of the Vallesian (late Miocene) in the Vallès–Penedès Basin (northeast Spain)

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Abstract

The magnetostratigraphic analysis of the late Miocene continental deposits from the Vallès–Penedès Basin, combined with its well-documented fossil mammal record, has provided a well-resolved chronology for the upper basin infill. The study is based on the biostratigraphic and magnetostratigraphic cross-correlation of 12 sections throughout the alluvial sequences in the western Vallès area. The biostratigraphic framework consists of 21 mammal localities corresponding to the Mammal Neogene MN9 and MN10 units. The composite magnetic polarity sequence is based on 400 paleomagnetic sites. Correlation with the Geomagnetic Polarity Time Scale (GPTS) has led to an absolute dating of the faunal events and a precise chronostratigraphy of the Vallesian mammal stage in its type area. The *Hipparion* First Appearance Datum (FAD), at the MN8/MN9 boundary, is dated at 11.1 Ma in the Vallès–Penedès Basin. This age, compared to other radiometrically dated localities in Europe, North Africa and Turkey, is consistent with an isochronous dispersal of this equid through the Mediterranean region. A possible isochrony at a larger geographical scale (Old World, Mesogea) must await more reliable ages of the *Hipparion* FAD in Asia and Africa. The *Cricetulodon* FAD (MN9a/MN9b boundary) is dated to 10.4 Ma, in chron C5n. The *Progonomys* FAD (MN9/MN10 boundary), corresponding to the intra-Vallesian faunal crisis, is dated at 9.7 Ma (C4Ar.3r). The Vallesian spans 2.4 Myr, from 11.1 Ma (C5r.1n) to 8.7 Ma (C4An) and correlates to the early Tortonian.

Keywords: Catalonia Spain; Vallesian; biostratigraphy; magnetostratigraphy

1. Introduction

In Tertiary continental sequences worldwide, the development of vertebrate paleontology has led to well-resolved biochronostratigraphic scales. Mammalian chronology is mainly founded on the rapid evolutionary patterns exhibited by rodent taxa combined with intercontinental migration events of large mammals. Although Neogene mammal biochronology has a high potential time resolution, mammal stages often lack a reliable calibration to the marine standard chronostratigraphy. In most basins, direct marine/continental faunal correlations rarely occur, while radiometric ages are often restricted to a few

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suitable layers.

Magnetostratigraphy can provide for continental sequences the necessary and precise age constraints and has led to advances in mammal biochronology in North America, Pakistan and other areas [1]. Despite the large number of paleontological sites in the Mediterranean region, the calibration of the mammal units has long shown little progress, mostly because many classical fossil localities were unsuitable for magnetostratigraphic purposes. However, recent progress on magnetostratigraphic dating in some Tertiary Spanish basins has notably improved the calibration of the Neogene mammal stages of western Europe, including the continent-equivalent Oligocene–Miocene boundary in the Ebro Basin [2,3], the Aragonian in the Teruel and Calatayud–Daroca basins [4], the Turolian in the Jucar–Cabriel Basin [5] and the late Pliocene in the Guadix–Baza Basin [6,7].

Among the Miocene continental stages, the Vallesian [8,9] in particular has lacked reliable absolute dating. Defined in 1948 to refer to the peculiar late Miocene faunal associations from the Vallès–Penedès Basin, the Vallesian stage quickly reached general acceptance in Europe and North Africa because of the simple criterion applied to its lower boundary, the appearance of the three-toed equid *Hipparion*, and its attributed isochrony throughout this region [10]. The isochrony of the *Hipparion* datum, however, was not firmly supported by data. On the contrary, further constraints of the FAD of *Hipparion* in the Mediterranean region [11,12] and Pakistan [13] made this hypothesis controversial [14]. Ages from various Vallesian localities in the Mediterranean have been poorly constrained as a consequence of unreliable radiometric dates [11], shortness or incompleteness of the magnetostratigraphic record [12], inaccurate location of mammal sites in the stratigraphic sequence [15,16] or ambiguous field correlations to marine beds [17,18]. Early magnetostratigraphic dates based on indirect correlations [19,20] frequently had to be revised in the light of new radiometric data or updated marine chronostratigraphy [21].

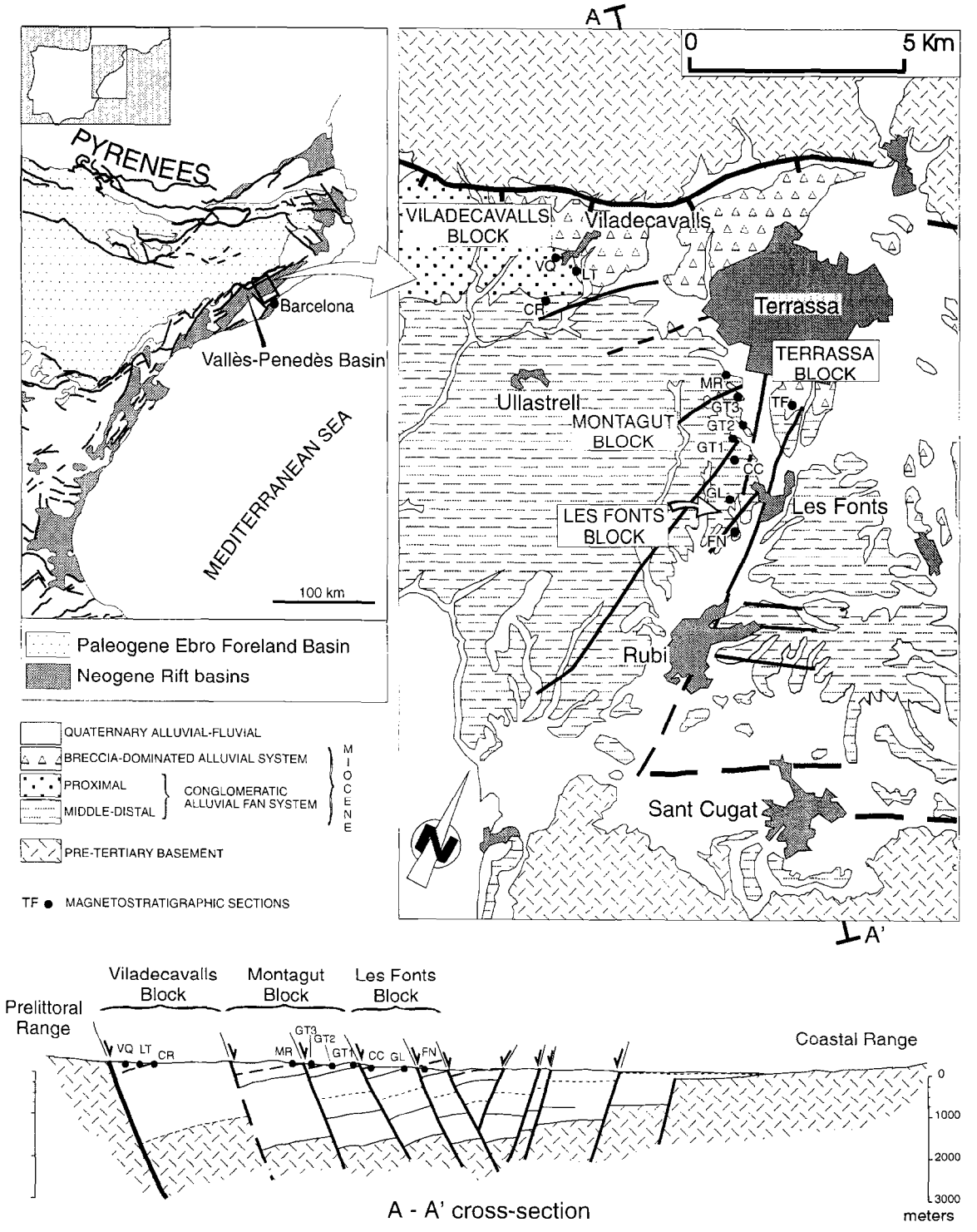
The main goal of the present study is to provide accurate ages of the mammal events that characterise the Vallesian stage in its type area. The results presented here are based on magnetostratigraphic analysis of 1300 m of combined alluvial sequences, including 21 fossil sites in a precisely known stratigraphic framework [22]. The composite magnetic polarity sequence is based on 400 paleomagnetic sites and allows an unambiguous and direct correlation of the Vallesian stratotype to the GPTS [23].

2. The Vallesian type area: geological setting

The Vallès–Penedès is a Neogene half-graben within the extensional structure of the Catalan margin of the Western Mediterranean Basin [24,25]. The basin infill consists of more than 2000 m of shallow marine to alluvial sediments, spanning the latest Oligocene to Pliocene. The upper half of the sedimentary infill records a prograding sequence from shallow marine deposits of Langhian age [26,27], through distal to middle–proximal alluvial deposits. During deposition of these units, tectonic subsidence was concentrated along the northwestern basin margin, while the southeastern passive margin recorded the overlap of sedimentary units prograding from the opposite margin and expanding over former southeastern margin faults.

The late Miocene alluvial sequences in the western Vallès consist of sediments from several concurrent alluvial systems that feed the basin from the northwestern active margin. Among these, the Olesa–Les Fonts alluvial fan contains the largest amount of sediment shed into the basin, radically spread over an area of more than 300 km^2. Its source area consisted of both pre-Tertiary basement (Paleozoic metamorphic rocks and Mesozoic red beds and marine limestones) and Ebro Basin Eocene sedimentary cover. Concurrent with the Olesa–Les Fonts

Fig. 1. Geological sketch and cross-section of the western Vallès area (Vallès–Penedès Basin) with localization of the magnetostratigraphic sections. *VQ* = Viladecavalls Quarry; *LT* = La Tarumba; *CR* = Can Coromines; *MR* = Les Martines; *GT3* = Can Guitart 3; *GT2* = Can Guitart 2; *GT1* = Can Guitart 1; *CC* = Creu de Conill; *GL* = Can Guilera; *CF* = Can Fonollet; *FN* = Torrent de Febulines.


A
0
5 Km
PYRENEES
Barcelona
Vallès-Penedès Basin
MEDITERRANEAN SEA
100 km
VILADECAVALLS BLOCK
Viladecavalls
VQ
LT
CR
Terrassa
TERRASSA BLOCK
MR
GT3
GT2
GT1
TF
Ullastrell
MONTAGUT BLOCK
CC
GL
FN
Les Fonts
LES FONTS BLOCK
Rubi
Sant Cugat
A'
Paleogene Ebro Foreland Basin
Neogene Rift basins
QUATERNARY ALLUVIAL-FLUVIAL
BRECCIA-DOMINATED ALLUVIAL SYSTEM
PROXIMAL
MIDDLE-DISTAL
CONGLOMERATIC ALLUVIAL FAN SYSTEM
MIOCENE
PRE-TERTIARY BASEMENT
TF ● MAGNETOSTRATIGRAPHIC SECTIONS
Viladecavalls Block
Montagut Block
Les Fonts Block
Prelittoral Range
Coastal Range
0
1000
2000
3000
meters
A - A' cross-section

conglomeratic fan, a breccia-dominated alluvial system developed along the hanging-wall side. This system consisted of a series of coalescing, small-scale fans, the evolution of which resulted from the tectonic activity of the main boundary fault. In the western Vallès, the breccia-dominated alluvial system is represented by the Terrassa–Viladecavalls fan. Its sediments consist of massive mudstones and mud-supported monogenic breccias that originated by erosion of the nearest massifs of the Prelittoral Range, with a Paleozoic metamorphic and volcanic source rock.

Miocene sedimentary sequences in western Vallès are predominantly tilted to the northwest by some 10–20° against the hanging wall in an overall roll-over like geometry (Fig. 1). The basin infill structure, dominated by intensive faulting along both the SW–NE primary direction and oblique N–S striking normal faults (Fig. 1), is the result of Miocene extension related to the rifting of the Valencia trough [24,28]. The surface fault distribution and geometry define a large number of small tilted blocks. The present study focuses on four blocks that delivered valuable biostratigraphic information and provided suitably thick sedimentary sequences for obtaining a composite biostratigraphic and magnetostratigraphic section.

3. Biostratigraphy: the Vallesian mammal units

The late Miocene alluvial sequences from the Vallès–Penedès Basin are widely known for their numerous and rich mammal fossil localities [29]. The peculiarities of the fossil assemblages (with middle Miocene affinities such as *Megacricetodon ibericus*) that appeared associated with the equid *Hipparion primigenium*, a late Miocene North American immigrant, led to the definition of the western Vallès as the type area of the Vallesian mammal stage [8,9]. The Vallesian is defined between the first appearance of *Hipparion* (i.e., the Aragonian/Vallesian boundary [30,31]) and the first appearance of the murid *Parapodemus lugdunensis* (i.e., the Vallesian/Turolian boundary [32]). Since its definition, the Vallesian has been subdivided into two main biochronological units, the early and the late Vallesian, which were later considered to be equivalent to

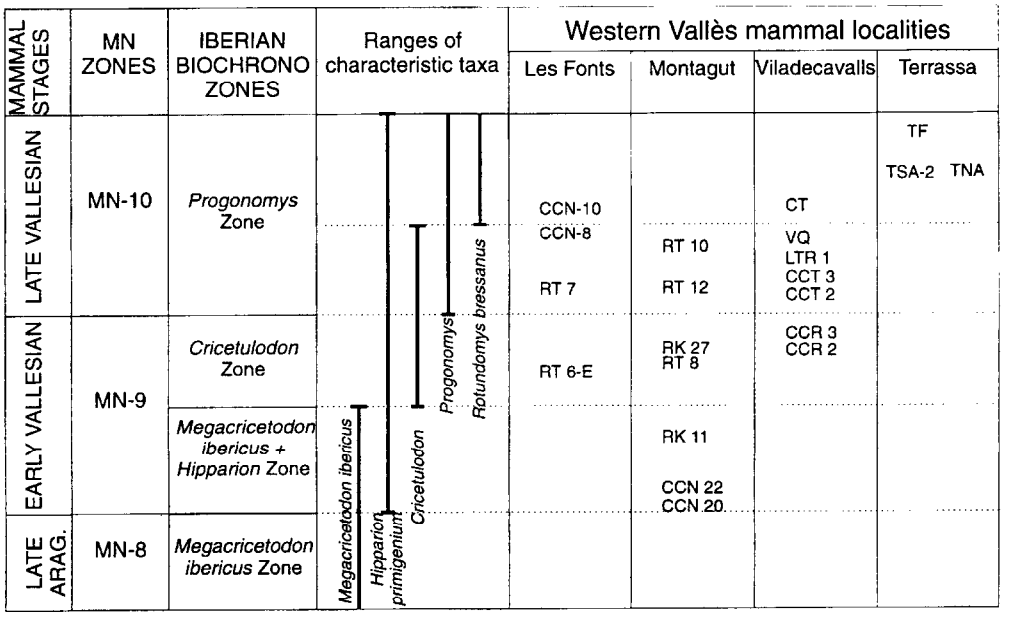


Fig. 2. The Vallesian mammal events recorded in the western Valles, their relationship with biochronological units [33] and mammal localities involved in this study. Classic known localities are: *VQ* = Viladecavalls Quarry; *LTR-1* = La Tarumba; *TNA* = Talud Norte Autopista; and *TF* = Torrent de Febulines.

the mammal units MN9 and MN10 [31], respectively.

The early Vallesian is characterised by the appearance of eastern immigrants such as *Hipparion* and *Cricetulodon*, to the pre-existing late Aragonian paleobiota. Despite its originally local significance, the attributed isochrony of the *Hipparion* datum [10] prompted the use of the Vallesian stage in other regions of Eurasia and North Africa.

The early/late Vallesian boundary is marked by one of the most distinct faunal events of the late Neogene: the dispersal of the murid *Progonomys* into Europe and North Africa. Dispersal of *Progonomys* in the western Mediterranean coincides with an important macromammal turnover [33], characterized by the exit of a large number of species and a decrease in diversity. Although the appearance of *Progonomys* is well documented in the Mediterranean region, no reliable ages of this event have so far been established. In the Vallès–Penedès Basin, *Progonomys* sp. is subsequently replaced by a murid of larger size, *Progonomys woelferi*, associated with the cricetid *Rotundomys bressanus*.

Paleontological research through the sections sampled for magnetostratigraphy provided a precise biostratigraphic framework, consisting of 17 new mammal sites of Vallesian age, in addition to the classical localities from Viladecavalls (La Tarumba and Viladecavalls Quarry) and Terrassa (Torrent de Febulines and Talud Norte Autopista) [29]. This data set outlines a sequence of faunal events (Fig. 2) which are in broad agreement with the formerly proposed biochronological zonation of the Neogene based on the study of Spanish basins [33]. The three biozones involved in the present study are the *Megacricetodon ibericus* + *Hipparion* Zone, the *Cricetulodon* Zone and the *Progonomys* Zone. The *Megacricetodon ibericus* + *Hipparion* Zone is defined by the appearance of *Hipparion*, an event that is not associated with any remarkable change in the rodent assemblages. The *Cricetulodon* Zone starts with the appearance of the immigrant cricetid *Cricetulodon* and the ursid *Indarctos*, among others. The late Vallesian *Progonomys* Zone is defined as between the first appearance of the murid *Progonomys* and the first appearance of *Parapodemus lugdunensis*, which defines the base of the Turolian stage. To date, *Parapodemus lugdunensis* has not been recorded in the Vallès–Penedès Basin. The upper part of the *Progonomys* Zone contains the first appearance of *Rotundomys bressanus*, another event that is recorded in some of the sections studied.

4. Sections sampled

The present magnetostratigraphic study comprises a total of 11 stratigraphic sections from four structural blocks. Biostratigraphic and magnetostratigraphic data, aided by mapping and field correlations, allowed us to build a composite section for each block. The Montagut and Les Fonts blocks (Fig. 3) are the ones that provide the most complete record of Vallesian faunas and will be used as reference sections for correlation of the other sections involved in this study.

The Montagut and Les Fonts composite sections are located in the inner part of the basin and are represented by the intermediate to marginal facies of the Olesa–Les Fonts alluvial fan (Fig. 1). Sediments consist of conglomeratic channel-fill sequences and mudstone overbank deposits with calcareous paleosoils. Conglomeratic sequences from the Montagut block interfinger with layers of massive breccias from the Terrassa–Viladecavalls fan. Biostratigraphic data from the Montagut and Les Fonts blocks come from newly discovered sites (Fig. 2). Biostratigraphic ages of sites in Montagut and Les Fonts span from the *Megacricetodon ibericus* + *Hipparion* Zone (lowermost early Vallesian) up to the *Progonomys* Zone (late Vallesian).

The Viladecavalls composite section is located along the northwestern margin of the basin (Fig. 1) and it is attached to the hanging wall, showing the more proximal and coarser sedimentary sequences. These consist of interfingering massive breccias from the Terrassa–Viladecavalls fan and conglomeratic facies from the Olesa–Les Fonts fan. The Viladecavalls area has traditionally yielded numerous mammal remains of late Vallesian age, as in the classical sites of Viladecavalls Quarry (VQ) and La Tarumba (LTR), now all integrated into the magnetostratigraphic sections of Viladecavalls Quarry and La Tarumba (Fig. 3). Our paleontological data have led to the recognition of the underlying early Vallesian fauna in the Can Coromines Section (sites CCR-2

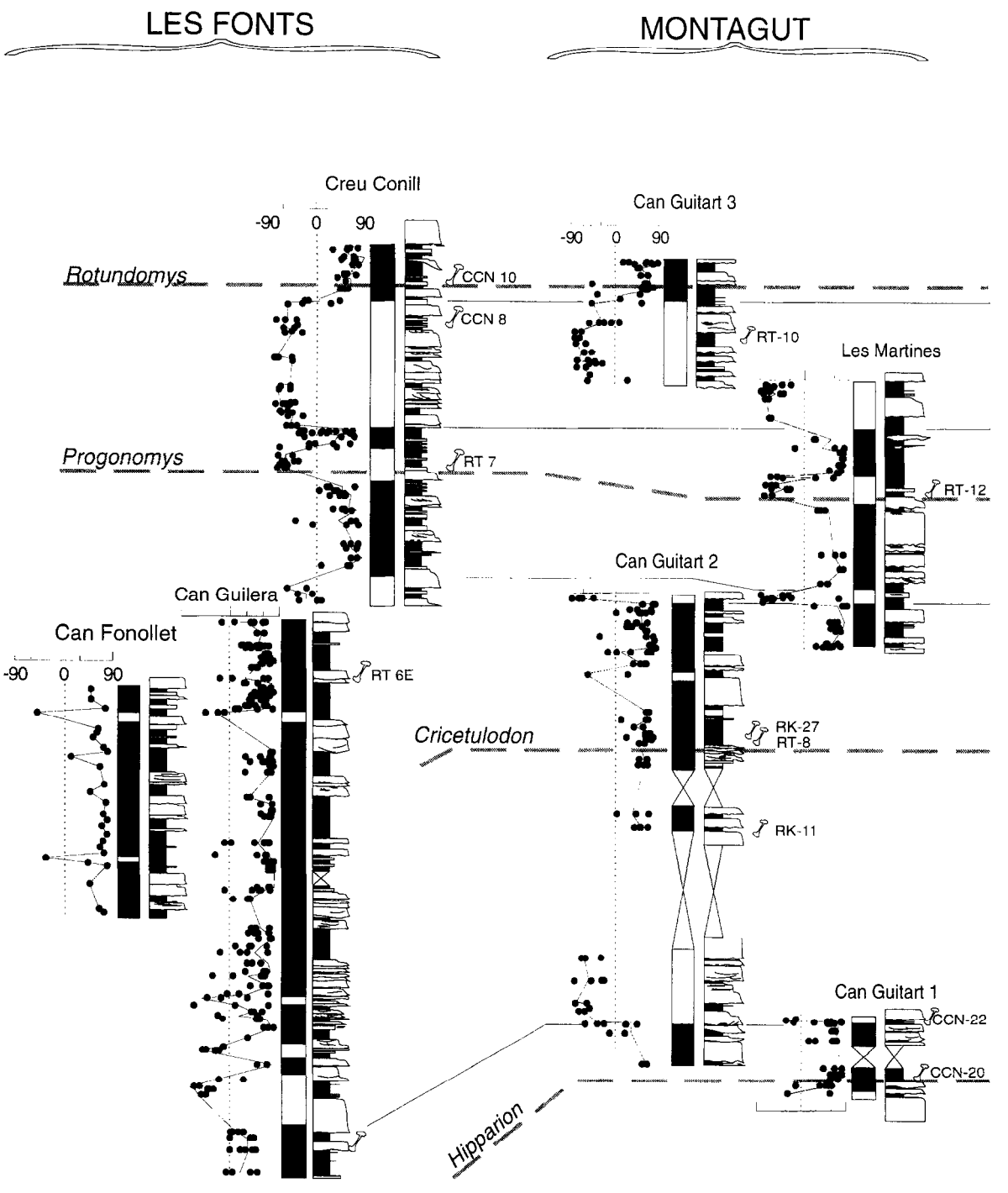


Fig. 3. Local magnetostratigraphic sections from (A) Les Fonts and Montagut and (B) Viladecavalls and Terrassa blocks. Positive and negative VGP latitudes (dots) represent normal and reversed polarities respectively. Magnetostratigraphic and biostratigraphic (dashed lines) correlations between all the sections.

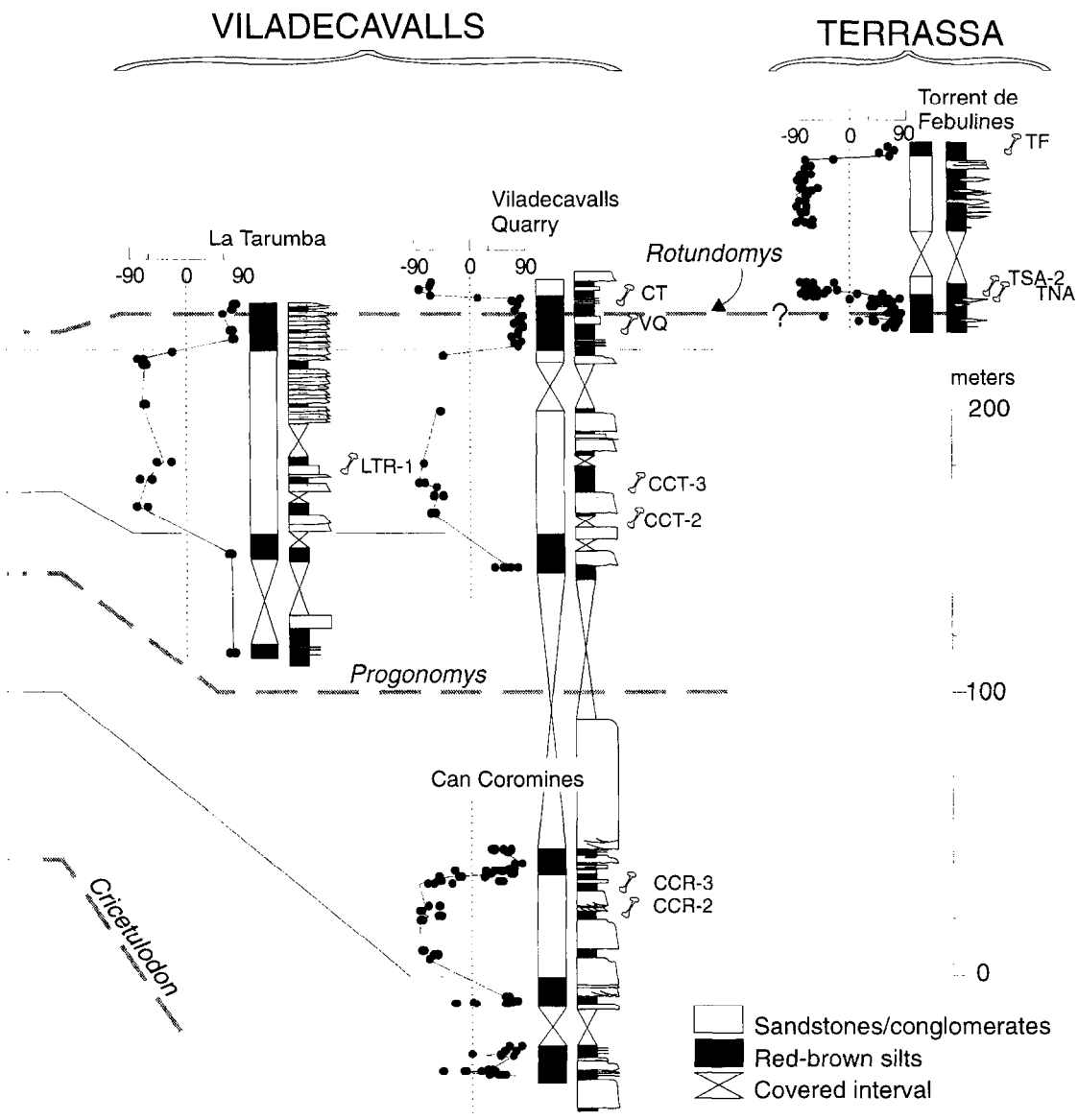


Fig. 3 (continued).

and CCR-3), as well as the overlying upper late Vallesian with *Rotundomys bressanus*, high in the Viladecavalls Quarry section (site CT).

The Terrassa composite section lies in a tectonically subsided region east of the area comprising the Viladecavalls, Montagut and Les Fonts blocks (Fig. 1). Sediments in the Terrassa block consist of massive muds and breccias from the Terrassa–Viladecavalls alluvial fan. Mammal remains are abundant in numerous classical sites, such as Torrent de Febulines, Talud Norte Autopista and Talud Sur Autopista (TF, TNA and TSA-2 in Figs. 2 and 3). All sites provided faunal associations of the *Progonomys* Zone with *Rotundomys bressanus* also present. The

presence of *Hipparion* cf. *mediterraneum* in the upper levels of the Terrassa section (TF) indicates an uppermost late Vallesian age.

5. Paleomagnetic analysis

Approximately 400 magnetostratigraphic levels were sampled from 1300 m of sedimentary sequences. The present paleomagnetic analysis is based on progressive thermal demagnetisation of three samples per stratigraphic level. Isothermal remanent magnetisation (IRM) analyses on samples from all sections and lithologies were also performed to understand the mineralogy of the magnetic carriers and the origin of the sediment magnetisation.

5.1. Magnetic properties and mineralogy

Magnetic properties of siliciclastic sediments are strongly dependent on the source rock, which, in turn, determines the mineralogy of detrital magnetic particles. Both the magnetic susceptibility and natural remanent magnetisation (NRM) of rocks are found to be influenced by the sediment source area. The magnetic susceptibility (K) of the sediments from the Terrassa–Viladecavalls alluvial fan is dominated by paramagnetic Fe-phyllosilicates and clay minerals [34], reflecting their volcanic and metamorphic source rock. Sediments from this unit show mean values of K well above 250×10^{-6} SI, twice as much as the average K in sediments from the Olesa–Les Fonts fan, with a more heterogeneous and less siliceous set of source rocks. Likewise, differences in NRM carriers between these two alluvial fans can be seen as a result of their distinctive source rocks. The NRM intensity in samples from the Terrassa–Viladecavalls fan (Fig. 4D) easily reached 10 mA/m, ten times higher than the ones from Olesa–Les Fonts (Fig. 4E). Stepwise thermal demagnetisation also reveals contrasting unblocking temperatures (T_b): approximately 600°C in Terrassa–Viladecavalls (Fig. 4D),

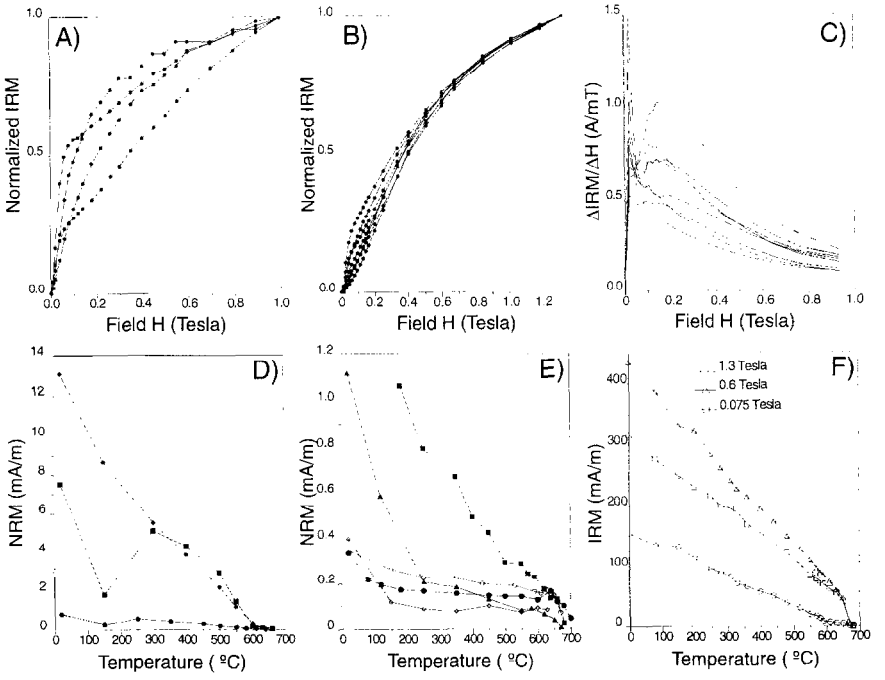


Fig. 4. Magnetic properties of sediments. IRM acquisition curves of samples from (A) the Terrassa–Viladecavalls and (B) the Olesa–Les Fonts (B). (C) Coercivity spectra of samples from the Olesa–Les Fonts alluvial fan. NRM thermal decay of samples from (D) the Terrassa–Viladecavalls and (E) the Olesa–Les Fonts alluvial fans. (F) IRM thermal decay showing the separate contribution of magnetite at the low-field curve and hematite at medium-and high-field curves.

but well above 650°C for samples from Olesa–Les Fonts (Fig. 4E). In both depositional systems, variable amounts of magnetite (or maghemite) are indicated by the steep slopes of the IRM acquisition curves at low fields (Fig. 4A,B), and peak values in the coercivity spectra (Fig. 4C) at fields as low as 0.03 T. This low coercivity fraction is dominant in the NRM of sediments from Terrassa–Viladecavalls, while in sediments from Olesa–Les Fonts the NRM is dominated by a larger hematite content (Fig. 4E). Separate demagnetisation of low (0.075 T), medium (0.6 T) and high field (1.3 T) IRM contributions [35] allowed identification of magnetite with a T_b of 580°C at the low-field IRM curve, and hematite with T_b of 675°C at the medium and high-field IRM curves (Fig. 4F).

The correlation between the ferromagnetic mineralogy and the sediment source rock suggests that detrital magnetic particles represent a large contribution to the total rock ferromagnetic content. The relatively large amounts of magnetite in sediments from the Terrassa–Viladecavalls alluvial fan are consistent with its magnetite-bearing metamorphic and volcanic source rock. Whether the hematite in sediments from Olesa–Les Fonts alluvial fan is of detrital origin or not is difficult to assess from the data. The significant inclination errors observed in the paleomagnetic directions would suggest, however, a detrital origin for most of the hematite [34,36].

5.2. Paleomagnetic directions and stability tests

For analysis of the NRM components, standard paleomagnetic procedures were followed. Three samples from each stratigraphic level were thermally demagnetised in steps of 50°C. Characteristic remanent magnetisation (ChRM) directions were calculated from the demagnetisation diagrams by means of principal component analysis [37]. Secondary viscous magnetisations conforming to the present-day field or acquired during sampling and/or storage were removed by heating to ~ 300°C. A characteristic component of either normal or reversed polarity was easily recognised above 400°C. Paleomagnetic directions were grouped into two classes. Class A directions (70% of the total) were obtained from demagnetisation plots displaying linear decays toward the origin (Fig. 5C–E). Class B directions

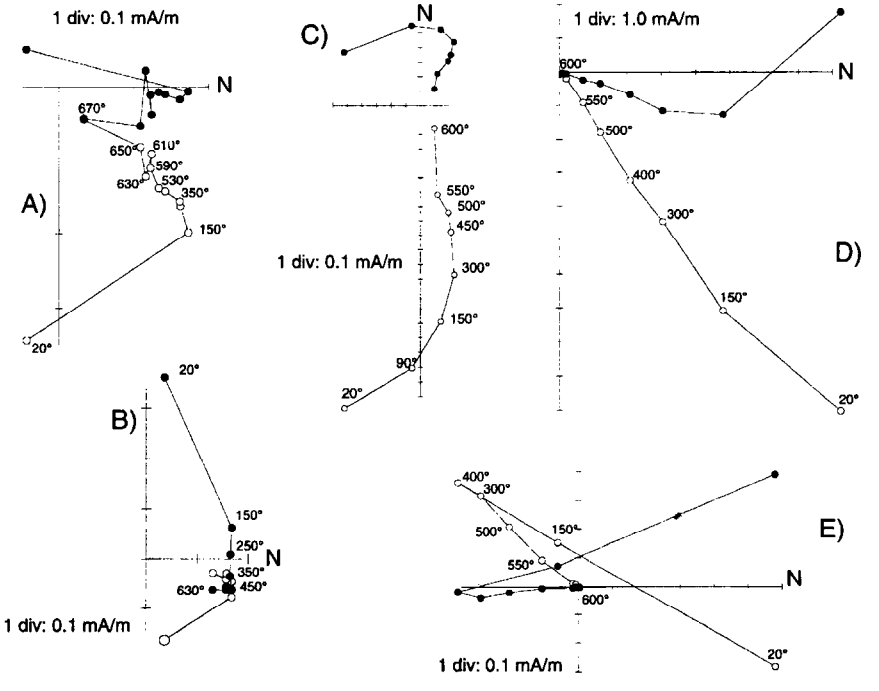


Fig. 5. Thermal demagnetisation diagrams of representative class A (C)–(E) and class B (A) and (B) samples.

(30% of the total) present demagnetisation trends not completed (Fig. 5A,B), with unstable directions probably caused by the addition of partial thermoremanence of newly formed minerals upon heating above 450°C. If complete demagnetisation was not obtained during thermal treatment, great circles provided additional information for estimating the mean site directions. Mean normal and reversed polarity directions of each magnetostratigraphic section (Fig. 6A) were calculated from the class A samples and great circles.

The reliability of magnetostratigraphic studies depends on the age constraints of NRM acquisition. In deformed sequences, fold tests may provide useful information about the age of magnetisation in relation to tectonics. Sediments from the Vallès–Penedès were not significantly deformed in the sites studied, and beds are gently and consistently tilted to the northwest. The prevalent tilt of sites makes a fold test [38] insignificant, but we note that the overall mean direction after tectonic correction is in agreement with an unrotated northerly reference direction (Fig. 6C), thus suggesting a pre-tilting magnetisation. The 16° clockwise rotation of the in situ mean declination relative to the northerly reference direction may well be explained as a result of an overall tilting against the northwest hanging wall.

A conglomerate test was applied to intraclasts (Fig. 6B) derived from within the alluvial system by erosion of semi-consolidated muddy floodplain sediments. The random distribution of ChRM directions derived from a sufficient number (N = 13) of boulders indicates that the ChRM is older than the deposition of the conglomerate. For intraformational conglomerates, the age of the ChRM is accurately constrained, since deposition of mud clasts takes place soon after re-working of the parent semi-consolidated mud sediments. The random distribution of

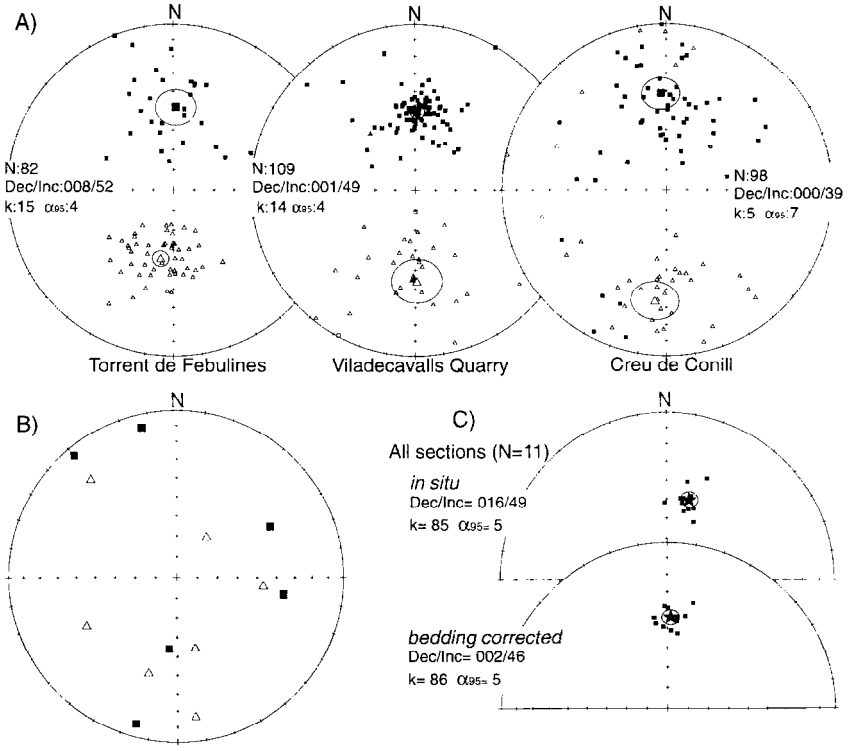


Fig. 6. (A) ChRM directions and stability tests. NRM directions from three magnetostratigraphic sections. (B) Positive conglomerate test carried out on intraformational mud clasts. (C) Stereographic projection of the site mean directions before and after tectonic correction. The similar bedding tilt of the sites makes a fold test insignificant, but the northerly directed overall mean direction after bedding correction suggests a pre-tilting magnetisation.

directions implies a positive conglomerate test, thus suggesting the presence of an early magnetisation in the sediments.

5.3. Magnetostratigraphy

The reliability of the magnetostratigraphic results is based not only on the field tests discussed above, but also on the consistency of the combined biostratigraphic and magnetostratigraphic cross-correlation throughout the sections (Fig. 3). The magnetic polarity of each stratigraphic level is represented by the virtual geomagnetic pole (VGP) latitude of the site mean direction. Correlation of the magnetic polarity sequences between the different structural blocks is consistent with the biostratigraphic constraints. The occurrence of very short magnetozones (as recorded by a single level) is considered unreliable, and was not taken into account for magnetostratigraphic correlation. Whether they represent true geomagnetic reversals or not is uncertain, and cannot be solved with more detailed sampling because of the prevalence of unsuitable, coarse-grained lithologies.

The pattern of magnetozones of the Les Fonts composite section indicates the occurrence of a long distinctive normal interval made up by the Can Fonollet and the Can Guilera sections (Fig. 3). The

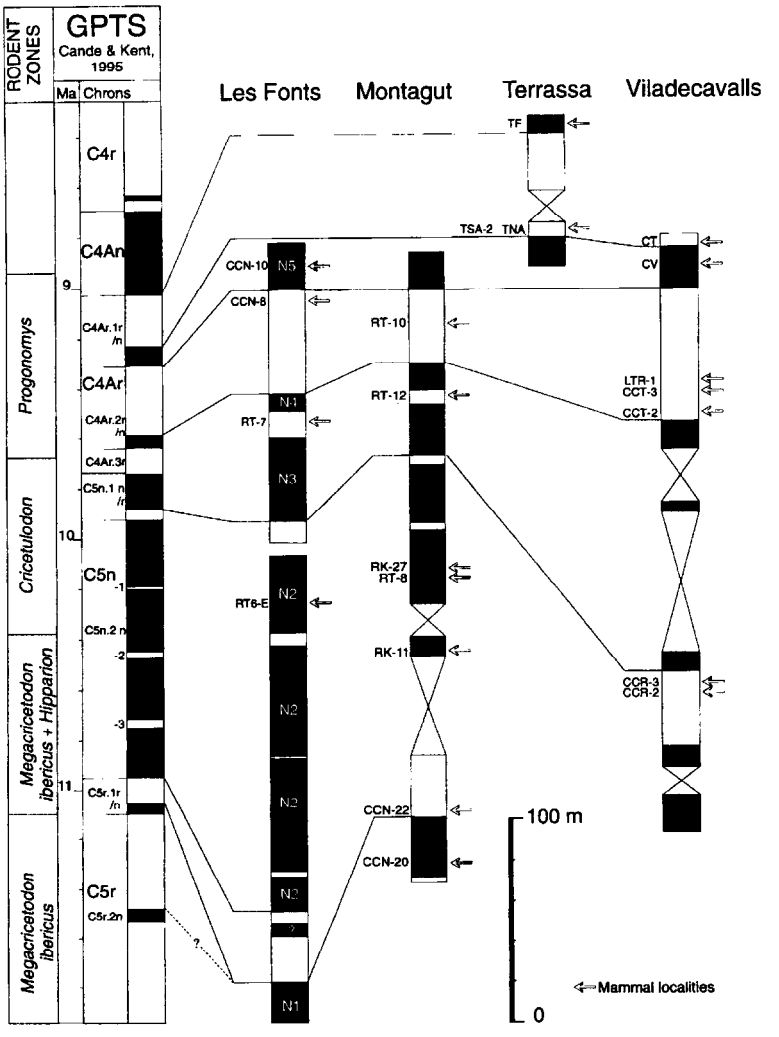


Fig. 7. Correlation of the Les Fonts, Montagut, Viladecavalls and Terrassa composite magnetic polarity sequences to the GPTS.

upper part of the Can Guilera section (site RT-6E) is dated as early Vallesian (*Cricetulodon* Zone), the same age as the two upper mammal sites from the Can Guitart 2 section (RT-8 and RK-27) in the Montagut block. These mammal sites also belong to a distinctive long normal magnetozone and allow the correlation between the Les Fonts and Montagut composite sections. The magnetic polarity pattern recorded in the late Vallesian Creu de Conill section (Les Fonts) agrees well with the sequence of reversals observed in the upper half of the Montagut composite section. On the basis of both the faunal assemblages and local magnetostratigraphy, we conclude that the correlation between the Montagut and the Les Fonts composites is unambiguous.

The vertebrate fossil record from the Viladecavalls block is known for its late Vallesian assemblages (*Progonomys* Zone). The recently discovered mammal site in the uppermost part of the Viladecavalls Quarry section (CT in Fig. 3) accurately records the first appearance of *Rotundomys bressanus*, a faunal event which is also recognised in the Les Fonts composite section. This biostratigraphic constraint allows a reliable magnetostratigraphic correlation between the Viladecavalls and Les Fonts/Montagut blocks (Fig. 3). As a result, the 300 m thick Viladecavalls composite section is time-equivalent to the uppermost 150 m from Les Fonts. Since the Viladecavalls composite section is located in the more subsiding basin margin, it has a much thicker stratigraphic sequence, consistent with the notable NW–SE trending half-graben asymmetry [24].

The Terrassa section holds the youngest faunal associations found in the western Vallès area. It records the presence of *Progonomys woelferi* in association with *Rotundomys bressanus*. However, its upper levels (site TF in Fig. 3) also record the appearance of *Hipparion* cf. *mediterraneum*. Based on the presence of *Rotundomys bressanus*, the lowermost normal magnetozone in the Terrassa section can be correlated to the uppermost normal magnetozones from both the Les Fonts and Viladecavalls composite sections. Although a younger age cannot be ruled out from the local magnetostratigraphy, the correlation of all the composite sections to the GPTS discussed below provides conclusive constraints to support this interpretation.

6. Correlation to the GPTS

The correlation of the sections studied to the GPTS is based on the long distinctive pattern of local magnetozones as well as on the recently established radiometric age of 9.6 ± 0.2 Ma for the late Vallesian faunal assemblages from the Bicorp Basin (eastern Spain) [39] and the known relationship of the Vallesian assemblages with the standard marine biozonations established in the Mediterranean. In the eastern Mediterranean Kastellios Hill sequence (Crete), similar late Vallesian faunas with *Hipparion*, *Cricetulodon* and *Progonomys* are associated with marine beds belonging to the N16 zone [40,41]. This N16 zone ranges from chron C5n to C4n [14]. After these constraints the correlation of the magnetostratigraphic sequences to the GPTS is best established by anchoring the distinctive long interval of normal polarity recorded in Les Fonts to Chron C5n (Fig. 7). The reversal sequence of our composite sections allows only one likely correlation to the GPTS, and it spans from the upper part of C5r to the lower part of C4An. According to the most recent calibration of the GPTS [23], the estimated ages for the Vallesian deposits in their type area are from 11.1 Ma to about 9.0 Ma. The correlation of the local magnetozone N1 (Fig. 7), at the base of the Les Fonts section, to the GPTS is not entirely certain. It could be either correlated with C5r.1n or C5r.2n. The ambiguity arises because the short normal event observed between N1 and N2 is based on a single level and was not considered significant for correlation. Because of the low reliability of such a short magnetozone, it is difficult to asses whether it represents a true geomagnetic feature. Correlation of N1 to C5r.1n is preferred as the most conservative (i.e. younger) for dating the lower boundary of the Vallesian stage.

The proposed correlation implies that the 400 m thick basin fill involved in this study displays smooth accumulation rates of, on average, 20 cm/kyr (Fig. 8). In the more tectonically active basin margin (Viladecavalls block), however, total subsidence is rated much higher, from 100 cm/kyr, in early Vallesian times, to 20 cm/kyr in the late Vallesian [22]. Despite the episodic character of alluvial sedimentation on a short time scale, smooth accumulation rates

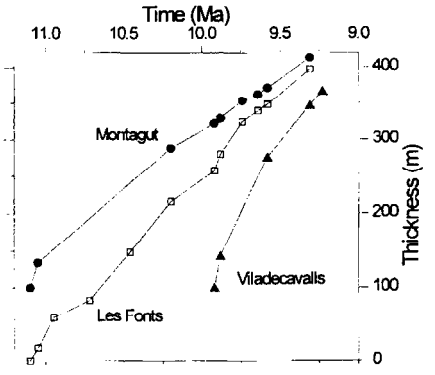


Fig. 8. Sediment accumulation rates derived from Les Fonts, Montagut and Viladecavalls composite sections.

are observed, implying that episodic sedimentation is averaged out over periods of 200 kyr (the mean duration of reversals for this time interval).

7. The chronology of the Vallesian

The present age determination of the Vallesian lower boundary relies on the correlation of local magnetozone N1 with C5r.1n (Fig. 7). This gives a minimum age of 11.1 Ma for the earliest known record of *Hipparion* in Vallès–Penedès. As discussed above, this age is the youngest of two possible correlations. The alternative correlation of N1 to C5r.2n, would imply that the Aragonian/Vallesian boundary is dated at 11.5 Ma. This age, however, is not well supported because of the lack of significance of the very short magnetozones at the base of the sequence, and should at present not be considered, unless it is confirmed by other studies. The FAD of *Cricetulodon* (lower boundary of *Cricetulodon* Zone) is dated at the middle part of C5n. Assuming an on average constant sediment accumulation rate, this datum has an age of approximately 10.4 Ma, based on both its position relative to the chron boundaries and the identification of cryptochron C5n.2n-1 20 m above the sites with the earliest record of *Cricetulodon* (site RT-8, Montagut block). The earliest appearance of *Progonomys* is found in chron C4Ar.3r, as recorded in both Les Fonts (site RT-7) and Montagut (site RT-12). The FAD of *Progonomys*, marks the transition from early to late Vallesian assemblages and is well recorded in the type locality of Can Llobateres, where magnetostratigraphic results accurately constrain its age to C4Ar.3r (9.6–9.7 Ma) [42]. This faunal event, which is coincident with a major faunal turnover in the Mediterranean region [33], has been shown to be remarkably heterochronous at a continental scale, since it is dated at about 12.5 Ma in the Potwar Plateau (Pakistan). Finally, the FAD of *Rotundomys bressanus* occurs in the upper part of C4Ar.1n (9.2–9.3 Ma), as recorded in Les Fonts (site CCN-10). The known age of the uppermost Vallesian record in Vallès–Penedès at about 9.0 Ma closely fits with the earliest Turolian faunas dated at 8.7 Ma in the Teruel Basin [43].

Our correlation agrees with the radiometric ages of the early *Hipparion* sites of Höwenegg in Germany (10.8 ± 0.3 Ma) [44] and the Vienna Basin (11.1 ± 0.5 Ma) [45], as well as with the Turkish localities of Yailacilar (11.6 ± 0.25 Ma) and Yenieskihisar (11.1 ± 0.2 Ma) where *Hipparion* is absent [46]. All these data support the isochrony of the dispersal of *Hipparion* throughout the Mediterranean region. The *Hipparion* sites of Bou Hanifia (Algeria) and Kastellios Hill (Crete), formerly correlated to chron C5r [12], need now to be reconsidered in the light of the improved marine–continental correlations in the Eastern Mediterranean [40,41] and cannot be considered to date the appearance of *Hipparion* in the Mediterranean. Based on the range of the N16 planktonic zone (from C5n to C4N), the magnetostratigraphic correlation of Kastellios Hill to chron C4Ar is preferred [21]. Furthermore, the association of *Hipparion* to *Progonomys* both in Bou Hanifia and Kastellios Hill also supports a correlation to chron C4Ar, in agreement with the age of the late Vallesian in the Vallès–Penedès Basin (this study).

Since the earliest recording of *Hipparion* in the Siwaliks (Pakistan) is found in the lower third of C5n ($\approx$ 10.6 Ma) [13], this faunal event might show a certain diachrony ($\approx$ 0.5 Myr) at a large geographical scale (Eurasia, Mesogea). It has been pointed out, however, that the appearance of *Hipparion* in the Siwaliks overlies a sequence of thick sandstone beds (the lower Nagri Formation) with a few mammal localities because of the unfavourable combination of unproductive lithologies and poor exposures [13,47]. The low resolution of the faunal record in

the lower Nagri Formation add some uncertainty to the observed diachrony of the datum in Siwaliks relative to the Mediterranean region. Still, provided such a diachrony is further confirmed, it cannot be as large as previously believed.

The appearance of *Hipparion* in Eurasia at about 11.1 Ma supports the dispersal of this equid from its North American ancestor *Cormohipparion occidentale* [48,49], the range of which is dated in the Dove Spring Formation (southern California) from about 12.6 Ma to 9.6 Ma [50], triggered by the pronounced late middle Miocene glacio-eustatic lowstand.

8. Conclusions

To provide mammal units with an appropriate chronostratigraphic meaning, reliable absolute dates for local mammal zones in their type and other areas are required. Interbasinal and intercontinental correlations must lead to determining the geographic area over which mammalian events can still be considered isochronous, and thus to establishing the spatial limits of biostratigraphic units as well as the timing and factors controlling terrestrial faunal dispersal. Combined biostratigraphic and magnetostratigraphic cross-correlation of different composite sections in the western Vallès (Vallès–Penedès Basin) have lead to a reliable chronostratigraphy of the Vallesian mammal stage in its type area. Correlation of the local polarity magnetic sequences to the GPTS has allowed dating of the sequence of bioevents and MN zones as recorded in the basin (Fig. 9). The age of the *Hipparion* FAD (the MN8/MN9 boundary) at 11.1 Ma is consistent with an isochronous dispersal of this equid throughout the Mediterranean region. The *Cricetulodon* FAD (the MN9a/MN9b boundary) is dated at 10.4 Ma, in chron C5n. The *Progonomys* FAD (the MN9/MN10 boundary), corresponding to the intra-Vallesian faunal crisis, is dated at 9.7 Ma (C4Ar.3r), exhibiting a marked heterochrony over Eurasia. The Vallesian in its type area spans 2.4 Myr, from 11.1 Ma (C5r.1n) to 8.7 Ma (C4An), and correlates to the early Tortonian (Fig. 7). These results agree well with other magnetostratigraphic studies in the Teruel and Duero basins in Spain [43].

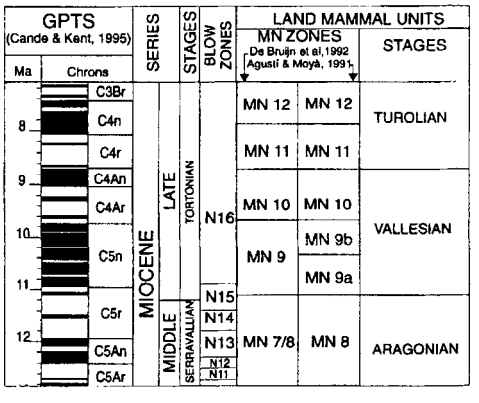


Fig. 9. The chronology of the Vallesian mammal stage and MN Zones. Correlation to the standard geochronologic time scale.

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